

CHAPTER TWO

KINEMATICS OF A PARTICLE: CURVILINEAR MOTION

Learning Objectives

After completing this chapter, students should be able to do the following:

- Analyse the kinematics of a particle undergoing curvilinear motion

2.1 General Curvilinear Motion

Curvilinear motion occurs when a particle moves along a **curved path**. Since this path is often described in **three dimensions**, **vector analysis** will be used to formulate the particle's position, velocity, and acceleration.

2.1.1 Position

Consider a particle located at a point on a space curve defined by the path function $s(t)$, Fig. 2.1a. The position of the particle, measured from a fixed point O , will be designated by the position vector $\mathbf{r} = \mathbf{r}(t)$. Notice that both the magnitude and direction of this vector will change as the particle moves along the curve.

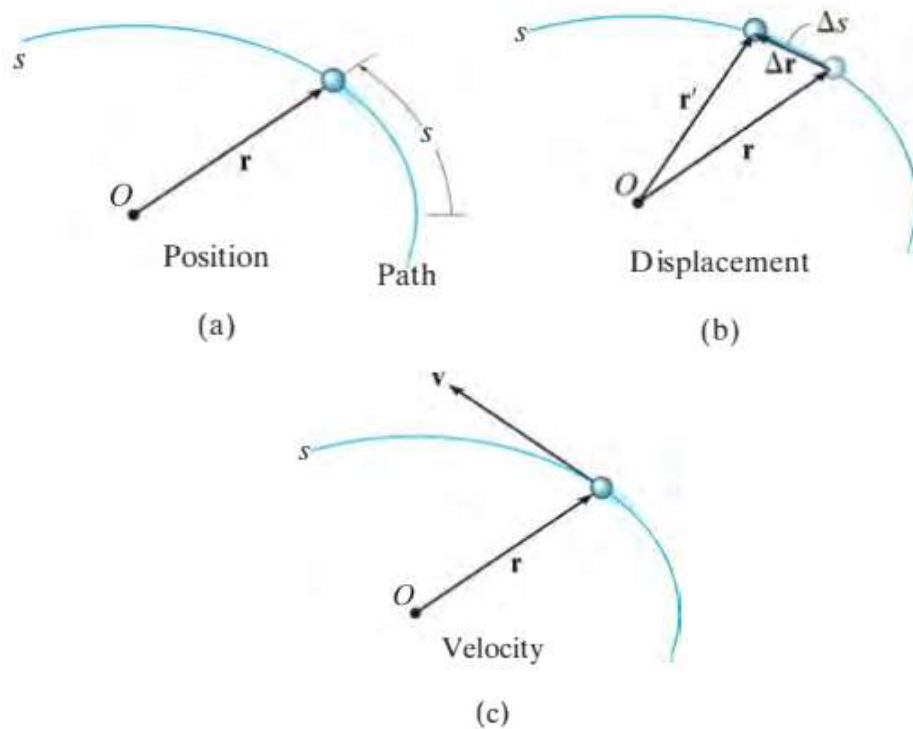


Fig. 2.1

2.1.2 Displacement

Suppose that during a small time interval Δt the particle moves a distance Δs along the curve to a new position, defined by $\mathbf{r}' = \mathbf{r} + \Delta \mathbf{r}$. Fig. 2.1b. The displacement $\Delta \mathbf{r}$ represents the change in the particle's position and is determined by vector subtraction; i.e., $\Delta \mathbf{r} = \mathbf{r}' - \mathbf{r}$.

2.1.3 Velocity

During the time Δt , the average velocity of the particle is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t} \quad (2.1)$$

The *instantaneous velocity* is determined from this equation by letting $\Delta t \rightarrow 0$, and consequently the direction of $\Delta \mathbf{r}$ approaches the *tangent* to the curve. Hence, $\mathbf{v} = \lim_{\Delta t \rightarrow 0} (\Delta \mathbf{r} / \Delta t)$ or

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} \quad (2.2)$$

Since $d\mathbf{r}$ will be tangent to the curve, the direction of $\mathbf{v}$ is also tangent to the curve, Fig. 2.1c. The magnitude of $\mathbf{v}$, which is called the speed, is obtained by realizing that the length of the straight line segment $\Delta \mathbf{r}$ in Fig. 2.1b approaches the arc length Δs as $\Delta t \rightarrow 0$, we have

$$v = \lim_{\Delta t \rightarrow 0} (\Delta r / \Delta t) = \lim_{\Delta t \rightarrow 0} (\Delta s / \Delta t), \text{ or}$$

$$v = \frac{ds}{dt} \quad (2.3)$$

Thus, the speed can be obtained by differentiating the path function s with respect to time.

2.1.4 Acceleration

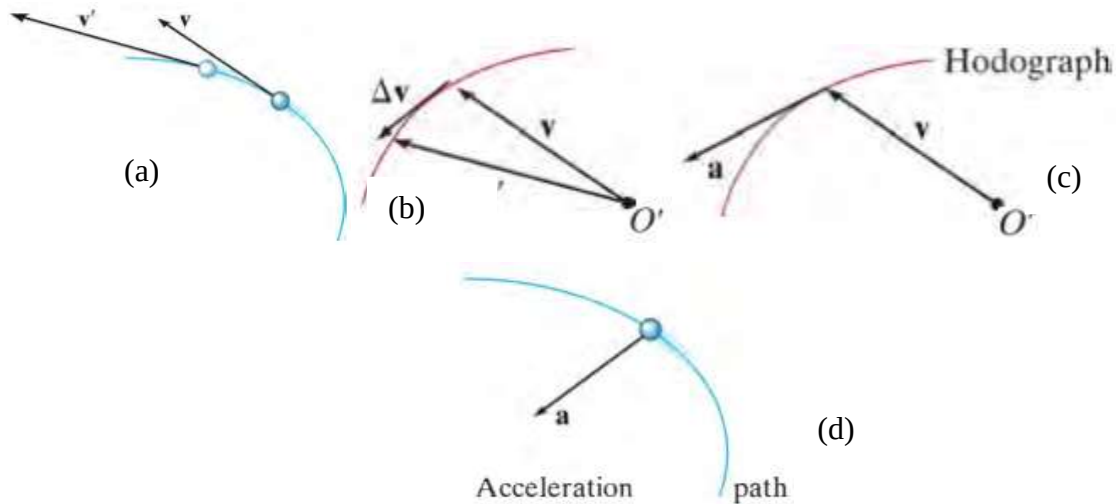


Fig. 2.2

If the particle has a velocity $\mathbf{v}$ at time t and a velocity $\mathbf{v}' = \mathbf{v} + \Delta\mathbf{v}$ at $t + \Delta t$, Fig. 2.2a, then the average acceleration of the particle during the time interval Δt is

$$\mathbf{a}_{avg} = \frac{\Delta\mathbf{v}}{\Delta t} \quad (2.4)$$

Where $\Delta\mathbf{v} = \mathbf{v}' - \mathbf{v}$

To obtain the instantaneous acceleration, let $\Delta t \rightarrow 0$ in the above equation. In the limit $\Delta\mathbf{v}$ will approach the tangent to the hodograph, and so $\mathbf{a} = \lim_{\Delta t \rightarrow 0} (\Delta\mathbf{v}/\Delta t)$, or

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} \quad (2.5)$$

Substituting Eq. 2.1 into this result, we can also write

$$\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2} \quad (2.6)$$

By definition of the derivative, $\mathbf{a}$ acts tangent to the hodograph, Fig. 2.2c, and, in general it is not tangent to the path of motion, Fig. 2.2d.

2.2 Curvilinear Motion: Rectangular Components

Occasionally the motion of a particle can best be described along a path that can be expressed in terms of its x, y, z coordinates.

2.2.1 Position

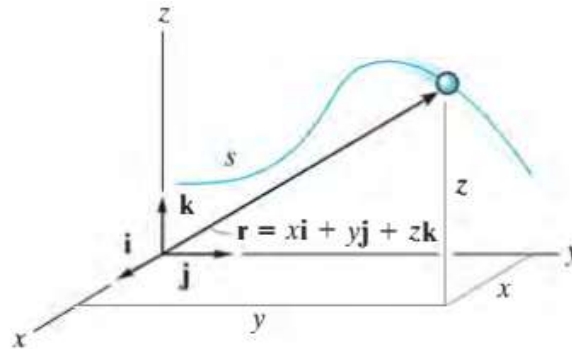


Fig. 2.3

Position

If the particle is at point (x, y, z) on the curved paths shown in Fig. 2.3, then its location is defined by the position vector

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (2.7)$$

When the particle moves, the x, y, z components of $\mathbf{r}$ will be functions of time; i.e., $x = x(t)$, $y = y(t)$, $z = z(t)$, so that $\mathbf{r} = \mathbf{r}(t)$.

At any instant the magnitude of $\mathbf{r}$ is defined as:

$$r = \sqrt{x^2 + y^2 + z^2} \quad (2.8)$$

And the direction of $\mathbf{r}$ is specified by the unit vector $\mathbf{u}_r = \mathbf{r}/r$.

2.2.2 Velocity

The first time derivative of $\mathbf{r}$ yields the velocity of the particle. Hence,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}(x\mathbf{i}) + \frac{d}{dt}(y\mathbf{j}) + \frac{d}{dt}(z\mathbf{k}) \quad (2.9)$$

When taking this derivative, it is necessary to account for changes in both the magnitude and direction of each of the vector's components. For example, the derivative of the $\mathbf{i}$ component of $\mathbf{r}$ is

$$\frac{d}{dt}(x\mathbf{i}) = \frac{dx}{dt}\mathbf{i} + x\frac{d\mathbf{i}}{dt} \quad (2.10)$$

The second term on the right side is zero, provided the x, y, z reference frame is fixed, and therefore the direction (and the magnitude) of $\mathbf{i}$ does not change with time. Differentiation of the $\mathbf{j}$ and $\mathbf{k}$ components may be carried out in a similar manner, which yields the final result,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k} \quad (2.11)$$

Where,

$$v_x = \dot{x}, v_y = \dot{y}, v_z = \dot{z}$$

The “dot” notation $\dot{x}, \dot{y}, \dot{z}$ represents the first time derivatives of $x = x(t), y = y(t), z = z(t)$, respectively.

The velocity has a *magnitude* that is found from

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (2.12)$$

and a direction that is specified by the unit vector $\mathbf{u}_v = \mathbf{v}/v$. As discussed in Section 2.1. This direction is always tangent to the path, as shown in Fig. 2.4.

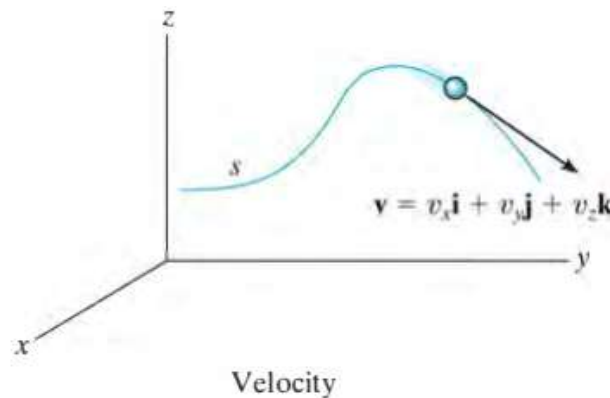


Fig. 2.4

2.2.3 Acceleration

The acceleration of the particle is obtained by taking the first time derivative of Eq. 2.11 (or the second time derivative of Eq. 2.7). We have

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k} \quad (2.13)$$

$$a_x = \dot{v}_x = \ddot{x} \quad (2.14)$$

$$a_y = \dot{v}_y = \ddot{y}$$

$$a_z = \dot{v}_z = \ddot{z}$$

Here a_x , a_y , a_z represent, respectively, the first time derivatives of $v_x = v_x(t)$, $v_y = v_y(t)$, $v_z = v_z(t)$, or the second time derivatives of the functions $x = x(t)$, $y = y(t)$, $z = z(t)$.

The acceleration has a magnitude

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad (2.15)$$

and a direction specified by the unit vector $\mathbf{u}_a = \mathbf{a}/a$. Since $\mathbf{a}$ represents the time rate of change in both the magnitude and direction of the velocity, in general $\mathbf{a}$ will not be tangent to the path, Fig. 2.5.

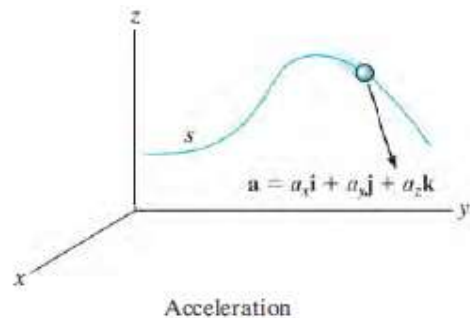
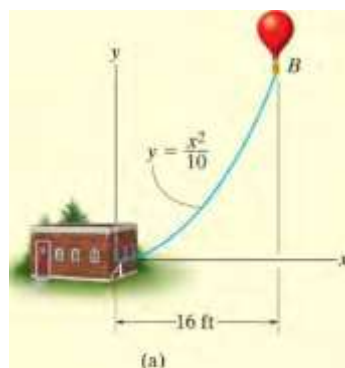


Fig. 2.5

Example 2.1

At any instant the horizontal position of the weather balloon in Fig. E-2.1a is defined by $x = (8t)ft$, where t is in seconds. If the equation of the path is $y = x^2/10$, determine the magnitude and direction of the (i) position, (ii) velocity and (iii) the acceleration when $t = 2$ s.



SOLUTION

Velocity. The velocity component in the x direction is

$$v_x = \dot{x} = \frac{d}{dt}(8t) = 8 \text{ ft/s} \rightarrow$$

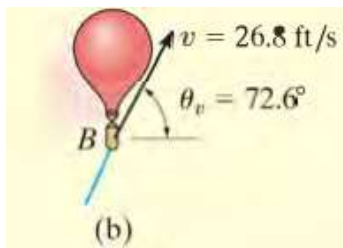
To find the relationship between the velocity components we will use the chain rule of calculus. (See Appendix A for a full explanation.)

$$v_y = \dot{y} = \frac{d}{dt}(x^2/10) = 2x\dot{x}/10 = 2(16)(8)/10 = 25.6 \text{ ft/s} \uparrow$$

When $t = 2$ s, the magnitude of velocity is therefore

$$v = \sqrt{(8 \text{ ft/s})^2 + (25.6 \text{ ft/s})^2} = 26.8 \text{ ft/s} \quad \text{Ans.}$$

The direction is tangent to the path, Fig. E-2.1b, where



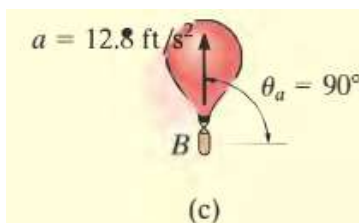
Acceleration. The relationship between the acceleration components is determined using the chain rule. (See Appendix C.) We have

$$\begin{aligned} a_x &= \dot{v}_x = \frac{d}{dt}(8) = 0 \\ a_y &= \dot{v}_y = \frac{d}{dt}(2x\dot{x}/10) = 2(\dot{x})\dot{x}/10 + 2x(\ddot{x})/10 \\ &= 2(8)^2/10 + 2(16)(0)/10 = 12.8 \text{ ft/s}^2 \uparrow \end{aligned}$$

Thus,

$$a = \sqrt{(0)^2 + (12.8)^2} = 12.8 \text{ ft/s}^2 \quad \text{Ans.}$$

The direction of $\mathbf{a}$, as shown in Fig. E-2.1c, is



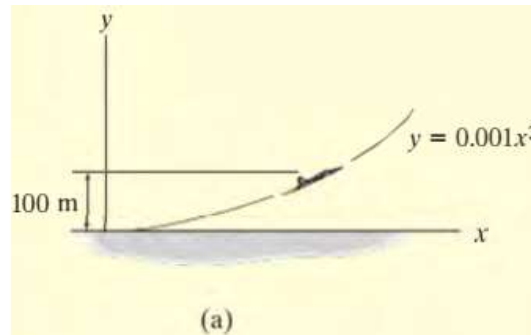
$$\theta_a = \tan^{-1} \frac{12.8}{0} = 90^\circ$$

Ans.

NOTE: It is also possible to obtain v_y and a_y by first expressing $y = f(t) = (8t)^2/10 = 6.4t^2$ and then taking successive time derivatives.

Example 2.2

For a short time, the path of the plane in Fig. E-2.2a is described by $y = (0.001x^2)$ m. If the plane is rising with a constant velocity of 10 m/s , determine the magnitudes of the (i) position, (ii) velocity and (ii) acceleration of the plane when it is at $y = 100$ m.



SOLUTION

When $y = 100$ m, then $100 = 0.001x^2$ or $x = 316.2$ m. Also, since $v_y = 10$ m/s, then

$$y = v_y t; \quad 100 \text{ m} = (10 \text{ m/s}) t \quad t = 10 \text{ s}$$

Velocity. Using the chain rule (see Appendix C) to find the relationship between the velocity components, we have

$$v_y = \dot{y} = \frac{d}{dt} (0.001x^2) = (0.002x)\dot{x} = 0.002xv_x \quad (1)$$

Thus

$$\begin{aligned} 10 \text{ m/s} &= 0.002(316.2 \text{ m})(v_x) \\ v_x &= 15.81 \text{ m/s} \end{aligned}$$

The magnitude of the velocity is therefore

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(15.81 \text{ m/s})^2 + (10 \text{ m/s})^2} = 18.7 \text{ m/s} \quad \text{Ans.}$$

Acceleration. Using the chain rule, the time derivative of Eq. (1) gives the relation between the acceleration components.

$$a_y = \dot{v}_y = 0.002\dot{x}v_x + 0.002x\dot{v}_x = 0.002(v_x^2 + xa_x)$$

When $x = 316.2 \text{ m}$, $v_x = 15.81 \text{ m/s}$, $\dot{v}_y = a_y = 0$,

$$0 = 0.002((15.81 \text{ m/s})^2 + 316.2 \text{ m}(a_x))$$

$$a_x = -0.791 \text{ m/s}^2$$

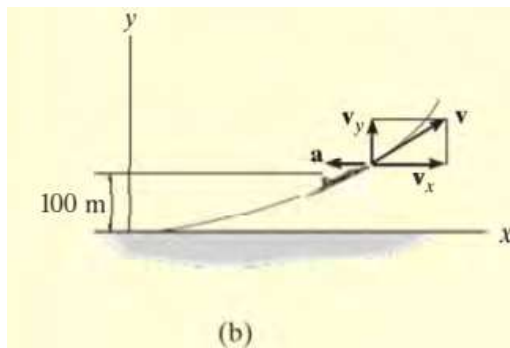
The magnitude of the plane's acceleration is therefore

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{(-0.791 \text{ m/s}^2)^2 + (0 \text{ m/s}^2)^2}$$

$$= 0.791 \text{ m/s}^2$$

Ans.

These results are shown in Fig. E-2.2b.



2.3 Motion of a Projectile

The free-flight motion of a projectile is often studied in terms of its rectangular components. To illustrate the kinematic analysis, consider a projectile launched at point (x_0, y_0) , with an initial velocity of $\mathbf{v}_0$, having components $(\mathbf{v}_0)_x$ and $(\mathbf{v}_0)_y$, Fig. 2.6.

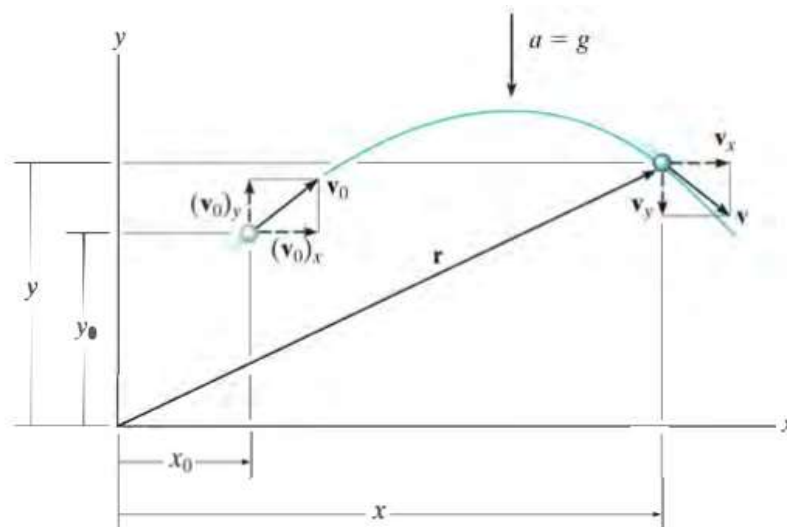


Fig. 2.6

When air resistance is neglected, the only force acting on the projectile is its weight, which causes the projectile to have a *constant downward acceleration* of approximately $a_c = g = 9.81 \text{ m/s}^2$ or $g = 32.2 \text{ ft/s}^2$.

2.3.1 Horizontal motion

Since $a_x = 0$, application of the constant acceleration equations, 1.4 to 1.6, yields

$$\begin{aligned} (\rightarrow) \quad v &= v_0 + a_c t; & v_x &= (v_0)_x \\ (\rightarrow) \quad x &= x_0 + v_0 t + \frac{1}{2} a_c t^2; & x &= x_0 + (v_0)_x t \\ (\rightarrow) \quad v^2 &= v_0^2 + 2a_c(x - x_0); & v_x &= (v_0)_x \end{aligned}$$

The first and last equations indicate that the horizontal component of velocity always remains constant during the motion.

2.3.2 Vertical motion

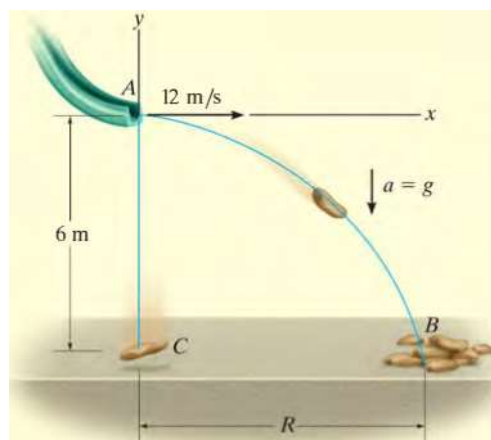
Since the positive y axis is directed upward, then $a_y = -g$. Applying Eqs. 1.4 to 1.6, we get

$$\begin{aligned} (+\uparrow) \quad v &= v_0 + a_c t; & v_y &= (v_0)_y - gt \\ (+\uparrow) \quad y &= y_0 + v_0 t + \frac{1}{2} a_c t^2; & y &= y_0 + (v_0)_y t - \frac{1}{2} gt^2 \\ (+\uparrow) \quad v^2 &= v_0^2 + 2a_c(y - y_0); & v_y^2 &= (v_0)_y^2 - 2g(y - y_0) \end{aligned}$$

Recall that the last equation can be formulated on the basis of eliminating the time t from the first two equations, and therefore only two of the above three equations are independent of one another.

Example 2.3

A sack slides off the ramp, shown in Fig. E-2.3, with a horizontal velocity of 12 m/s. If the height of the ramp is 6 m from the floor, determine the time needed for the sack to strike the floor and the range R where sacks begin to pile up.



SOLUTION

Coordinate System. The origin of coordinates is established at the beginning of the path, point A, Fig. E-2.3. The initial velocity of a sack has

components $(v_A)_x = 12 \text{ m/s}$ and $(v_A)_y = 0$. Also, between points A and B the acceleration is $a_y = -9.81 \text{ m/s}^2$. Since $(v_B)_x = (v_A)_x = 12 \text{ m/s}$, the three unknowns are $(v_B)_y$, R , and the time of flight t_{AB} . Here we do not need to determine $(v_B)_y$.

Vertical Motion. The vertical distance from A to B is known, and therefore we can obtain a direct solution for t_{AB} by using the equation

$$\begin{aligned} (+\uparrow) \quad y_B &= y_A + (v_A)_y t_{AB} + \frac{1}{2} a_c t_{AB}^2 \\ -6 \text{ m} &= 0 + 0 + \frac{1}{2} (-9.81 \text{ m/s}^2) t_{AB}^2 \\ t_{AB} &= 1.11 \text{ s} \end{aligned} \quad \text{Ans.}$$

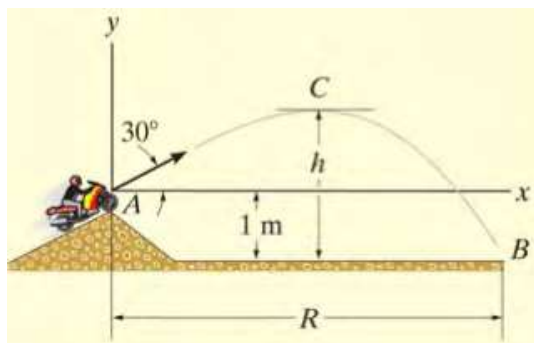
Horizontal Motion. Since t_{AB} has been calculated, R is determined as follows:

$$\begin{aligned} (\rightarrow) \quad x_B &= x_A + (v_A)_x t_{AB} \\ R &= 0 + 12 \text{ m/s} (1.11 \text{ s}) \\ R &= 13.3 \text{ m} \end{aligned} \quad \text{Ans.}$$

NOTE: The calculation for t_{AB} also indicates that if a sack were released from rest at A, it would take the same amount of time to strike the floor at C, Fig. E-2.3.

Example 2.4

The track for this racing event was designed so that riders jump off the slope at 30° , from a height of 1 m. During a race it was observed that the rider shown in Fig. E-2.3 remained in mid-air for 1.5s. Determine the speed at which he was traveling off the ramp, the horizontal distance he travels before striking the ground, and the maximum height he attains. Neglect the size of the bike and rider.



SOLUTION

Coordinate System. As shown in Fig. E-2.3, the origin of the coordinates is established at A. Between the end points of the path AB the three unknowns are the initial speed v_A , range R , and the vertical component of velocity $(v_B)_y$

Vertical Motion. Since the time of flight and the vertical distance between the ends of the path are known, we can determine v_A .

$$\begin{aligned}
 (+\uparrow) \quad y_B &= y_A + (v_A)_y t_{AB} + \frac{1}{2} a_c t_{AB}^2 \\
 -1 \text{ m} &= 0 + v_A \sin 30^\circ (1.5 \text{ s}) + \frac{1}{2} (-9.81 \text{ m/s}^2) (1.5 \text{ s})^2 \\
 v_A &= 13.38 \text{ m/s} = 13.4 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

Horizontal Motion. The range R can now be determined.

$$\begin{aligned}
 (\rightarrow) \quad x_B &= x_A + (v_A)_x t_{AB} \\
 R &= 0 + 13.38 \cos 30^\circ \text{ m/s} (1.5 \text{ s}) \\
 &= 17.4 \text{ m} \quad \text{Ans.}
 \end{aligned}$$

In order to find the maximum height h we will consider the path AC, Fig. E-2.3. Here the three unknowns are the time of flight t_{AC} , the horizontal distance from A to C, and the height h . At the maximum height $(v_C)_y = 0$, and since v_A is known, we can determine h directly without considering t_{AC} using the following equation.

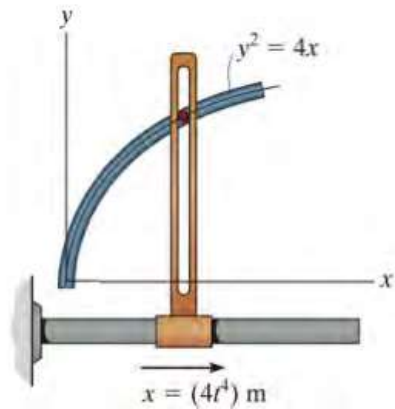
$$\begin{aligned}
 (v_C)_y^2 &= (v_A)_y^2 + 2a_c[y_C - y_A] \\
 0^2 &= (13.38 \sin 30^\circ \text{ m/s})^2 + 2(-9.81 \text{ m/s}^2)[(h - 1 \text{ m}) - 0] \\
 h &= 3.28 \text{ m} \quad \text{Ans.}
 \end{aligned}$$

NOTE: Show that the bike will strike the ground at B with a velocity having components of

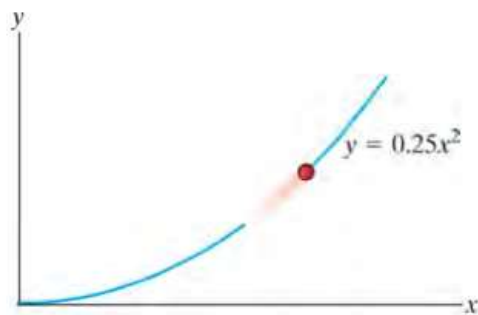
$$(v_B)_x = 11.6 \text{ m/s} \rightarrow, \quad (v_B)_y = 8.02 \text{ m/s} \downarrow$$

Problem Set 2.1

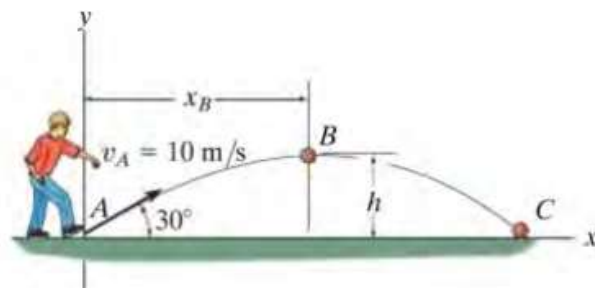
1. A particle is constrained to travel along the path shown in Figure below. If $x = (4t^2)m$, where t is in seconds, determine the magnitude of the particle's velocity and acceleration when $t = 0.5$ s



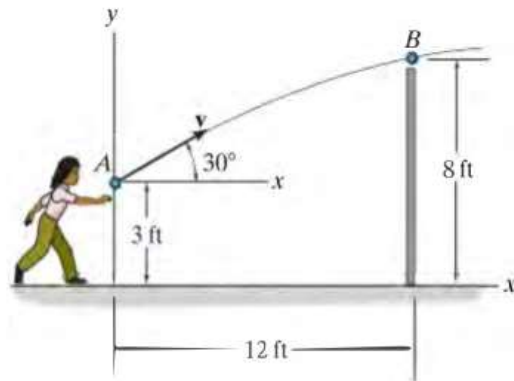
2. A particle is traveling along the parabolic path $y = 0.25x^2$ as shown in Figure below. If $x = (2t^2)m$, where t is in seconds, determine the magnitude of the particle's velocity and acceleration when $t = 2$ s.



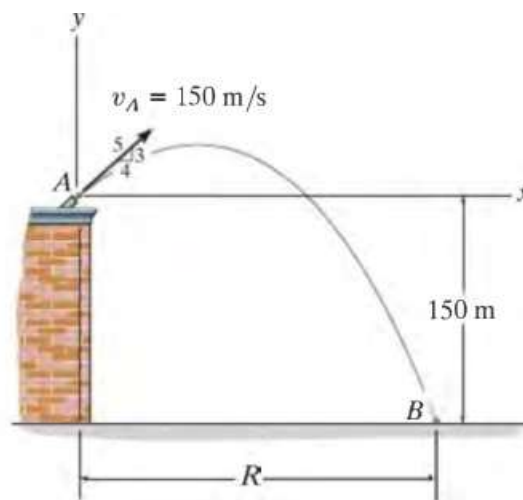
3. The ball is kicked from point A with the initial velocity $v_A = 10 \text{ m/s}$. Determine the range R , and the speed when the ball strikes the ground as shown in Figure below.



4. A ball is thrown from A shown in Figure below. If it is required to clear the wall at B, determine the minimum magnitude of its initial velocity v_A in m/s . [Note: $1\text{ft} = 0.3 \text{ m}$]



5. A projectile is fired with an initial velocity of $v_A = 150 \text{ m/s}$ off the roof of the building. Determine the range R where it strikes the ground at B as shown in Figure below.



6. The position of a particle is $\mathbf{r} = \{(3t^3 - 2t)\mathbf{i} - (4t^{\frac{1}{2}} + t)\mathbf{j} + (3t^2 - 2)\mathbf{k}\} \text{ m}$, where t is in seconds. Determine the magnitude of the particle's velocity and acceleration when $t = 2 \text{ s}$.
7. The velocity of a particle is given by $\mathbf{v} = \{16t^2\mathbf{i} - 4t^3\mathbf{j} + (5t + 2)\mathbf{k}\} \text{ m/s}$, where t is in seconds. If the particle is at the origin when $t = 0$, determine the magnitude of the particle's acceleration when $t = 2 \text{ s}$. Also, what is the x, y, z coordinate position of the particle at this instant?

